

LECTURE 15

EER Modeling: A Case Study

From client requirements to a complete, validated EER diagram

CSC270 · Database Systems

COMSATS University Islamabad, Lahore Campus

In This Lecture

- 1 The case study: Al-Shifa General Hospital
- 2 Entity types, relationships, and attributes — each decision justified
- 3 Weak entities, specialization levels, composition, and aggregation
- 4 Building the diagram one decision at a time
- 5 Validating the model against every requirement

Al-Shifa General Hospital

- A private hospital replaces its paper records. The administrator's brief, reduced to its facts:

Patients

Every patient, every admission (ward, bed, dates) and every out-patient visit.

Staff

Doctors, nurses, lab technicians, administrators. Each in one department. Doctors are consultants or residents.

Structure

Departments with a head doctor. Wards inside departments. Beds inside wards.

Care and billing

Attending doctor per admission. Appointments, prescriptions, invoices, and payment status.

How to Read a Requirements Brief

- 1 Entities** Nouns with their own attributes, relationships, or existence.
- 2 Relationships** Verb phrases between entities, with cardinality.
- 3 Attributes** Facts that describe exactly one entity.
- 4 Constraints** Specialization, weak entities, composition, aggregation.

THE DISCIPLINE

Check every decision against the sentence that motivated it.

STEP 1

Identifying Entity Types

Candidate	Entity?	Why
Patient	Yes	Own attributes; many relationships
Staff	Yes (superclass)	Shared by every kind of employee
Doctor, Nurse, Technician, Admin	Yes (subclasses)	Role-specific attributes
Consultant, Resident	Yes (2nd level)	Attributes the other lacks
Department, Ward	Yes	Own name, location, head; ward owns beds
Bed	Yes — weak	Cannot be identified without its ward
Admission, Appointment, Prescription	Yes	Own dates, numbers, dosages
Drug, Invoice	Yes	Independent catalog; own lifecycle
Consultation fee, CNIC	No — attribute	Single values for one entity

STEP 2

Identifying Relationships

Relationship	Participants	Cardinality	Note
WorksIn	Staff, Department	N:1	Exactly one department
HeadedBy	Department, Doctor	1:1	One head per department
Supervises	Consultant, Resident	1:N	Unary, via Doctor
ComposedOf	Department, Ward	1:N	Composition
Owns	Ward, Bed	1:N	Bed is weak
AdmittedTo	Patient, Bed	N:1	Carries dates → Admission
AttendedBy	Admission, Doctor	N:1	Attending doctor per stay
Books, ScheduledWith	Patient, Appointment, Doctor	1:N, N:1	Appointment per doctor
Prescribes, For, Of	Doctor, Prescription, Patient, Drug	1:N, N:1, N:1	Three-way tie
BilledTo	Patient, Invoice	1:N	Several invoices over time

STEP 3

Identifying Attributes

Patient

patientNo

firstName

dateOfBirth

phoneNumbers {}

Staff (before specialization)

staffNo

cnic

name

dateJoined

deptNo

WHAT IS LEFT OUT

age is NOT stored: it is derived from dateOfBirth (Lecture 13). phoneNumbers is multi-valued, so it becomes its own relation later.

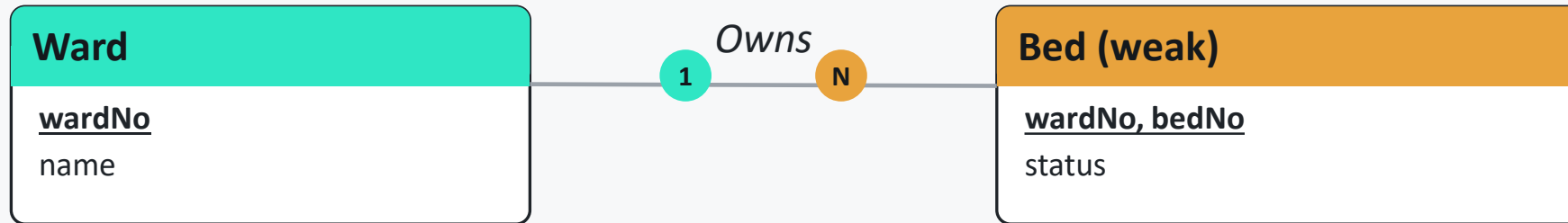
THE TEST

Each entity's remaining attributes follow the same test: keep it on the entity it describes, and nothing else.

STEP 4

Bed: A Weak Entity

- Bed 3 in Ward W1 is not Bed 3 in Ward W2. bedNo alone cannot identify a bed.



THE WEAK-ENTITY RULE

Bed's key is its partial key (bedNo) plus the owner's entire key (wardNo). Deleting a ward deletes its beds.

Why Build Layer by Layer, in This Order

- Start with the hub entity every question touches: Patient.
- Add one relationship at a time, checking each against the brief.
- Attach the specialization and the composition chains once their entities exist.
- Fuse everything last — a fan or chasm trap is far easier to see in a small diagram.

Patient

Appointment

Doctor

Staff hierarchy

Ward and Bed

Admission

Prescription

Invoice

Start With the Hub: Patient

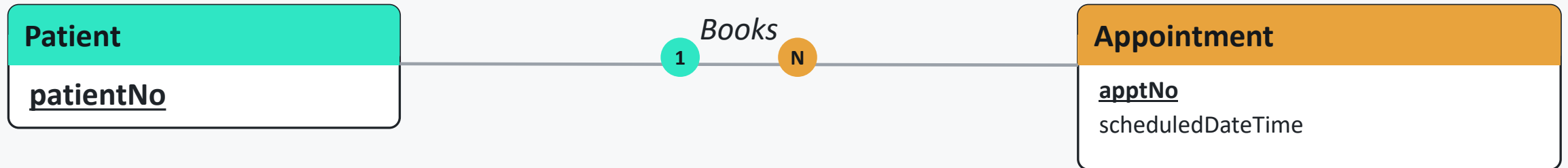
Patient

patientNo

WHY PATIENT FIRST

Reason: the brief says “we need to keep track of every patient.” Patient has an independent existence and is the entity most other facts hang from.

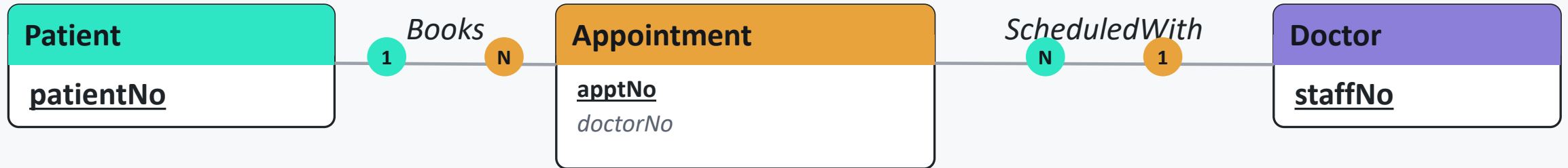
Add Appointments: the Books Relationship



WHY APPOINTMENT IS AN ENTITY

Reason: the appointment needs its own confirmation number, so it is an entity. A patient books many over time: 1:N.

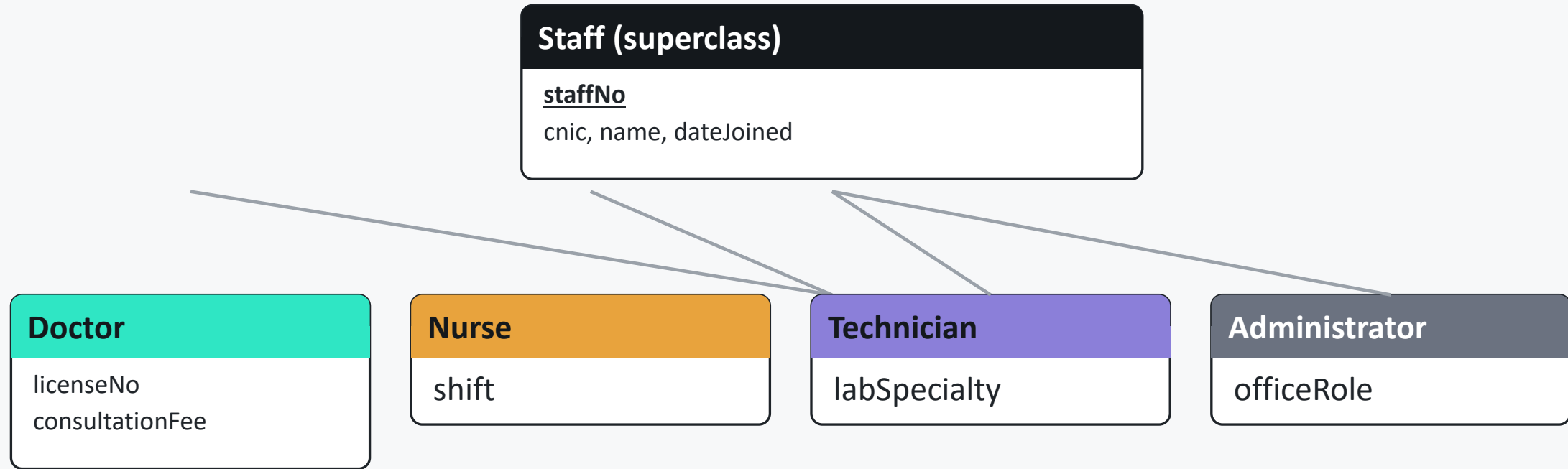
Complete Layer 1: Add the Doctor



WHERE THE FOREIGN KEY GOES

Each appointment is with exactly one doctor, so the foreign key sits on Appointment (the N side).

Staff Into Four Job Roles

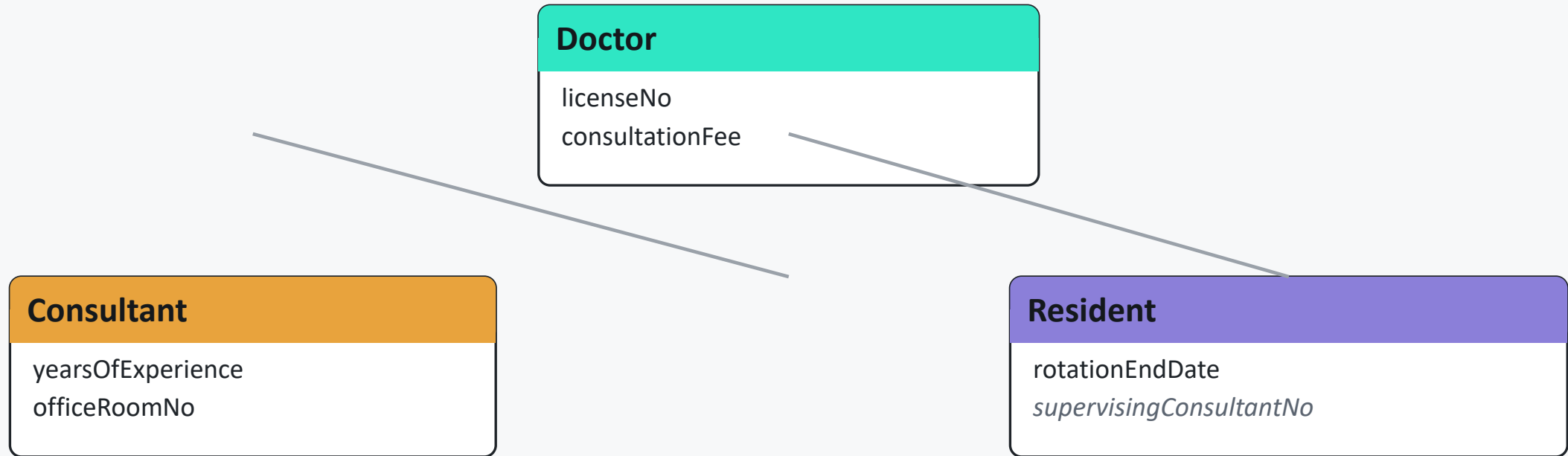


d DISJOINT, TOTAL

Disjoint (no one holds two roles) and total (every staff member holds one of the four).

Doctor Into Consultant and Resident

- “Some of our doctors are consultants ... the rest are residents”: a second level, under Doctor.



d DISJOINT, TOTAL

Disjoint and total. Resident.supervisingConsultantNo is a foreign key into Consultant: only consultants supervise.

Why Two Specialization Levels?

- A flat list of six leaves under Staff would duplicate licenseNo and consultationFee onto Consultant and Resident.
- Shared structure belongs at the highest level it applies to: every doctor has a license and a fee.
- Check each node: attributes shared by only some children mean a missing intermediate level.

Inheritance in This Hierarchy

Every Doctor, Consultant, and Resident

inherits staffNo, cnic, name, dateJoined, deptNo from Staff, plus licenseNo and consultationFee from Doctor.

WorksIn applies to everyone

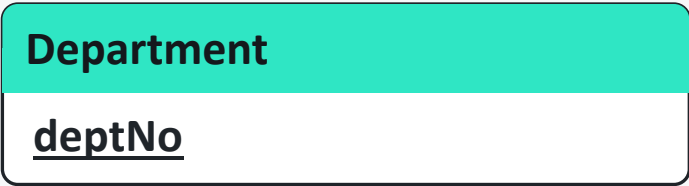
a Consultant works in a department exactly as a Nurse does, with no extra drawing.

HeadedBy and Supervises

belong to Doctor, Consultant, and Resident only. A Nurse never participates in either.

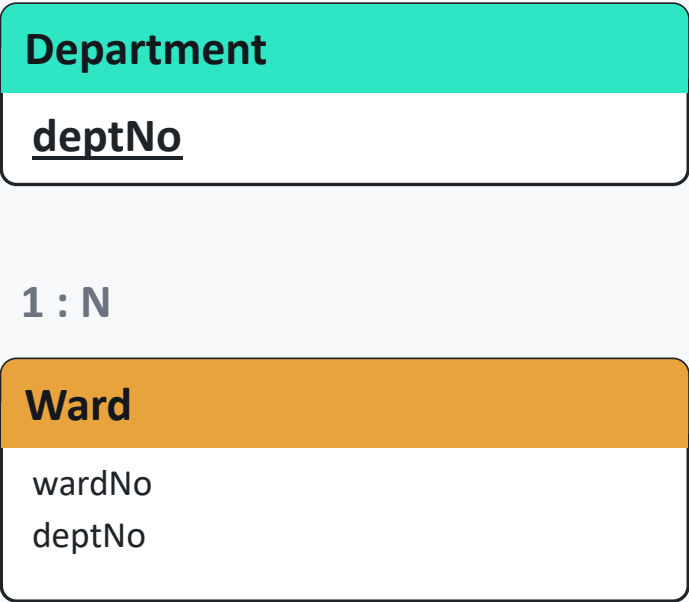
Growing Department → Ward → Bed

a) Department alone



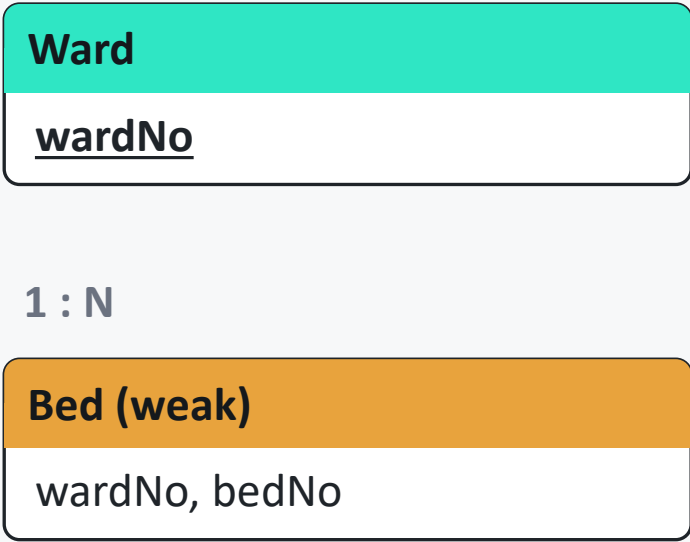
The anchor. Wards exist only inside a department.

b) Add Ward (composition)



A ward cannot exist outside its department: composition.

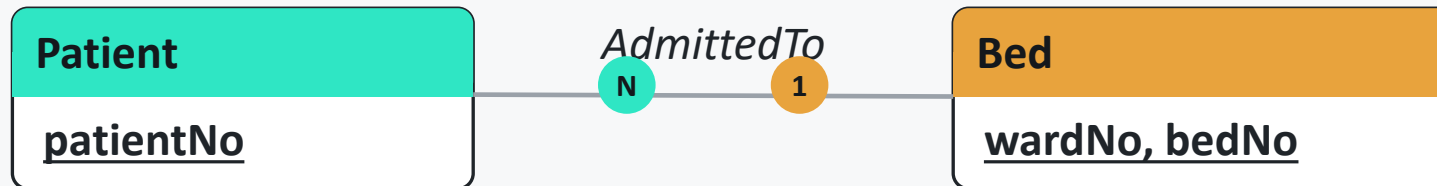
c) Add Bed (weak)



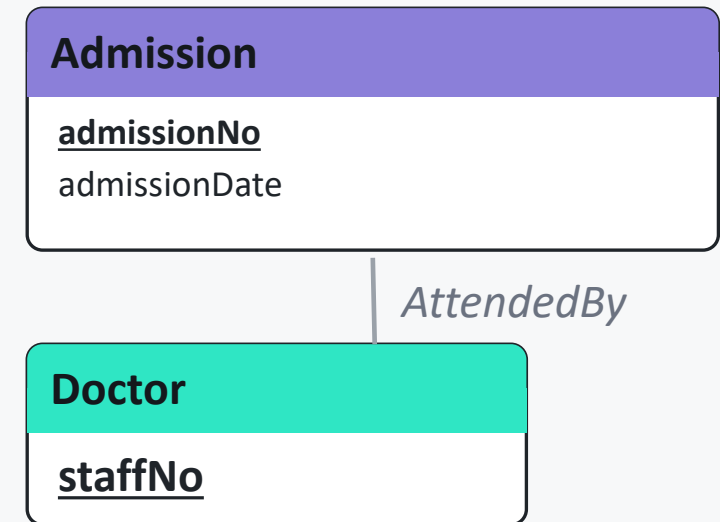
A bed is identified only through its ward.

AdmittedTo Becomes Admission

Before — a plain relationship



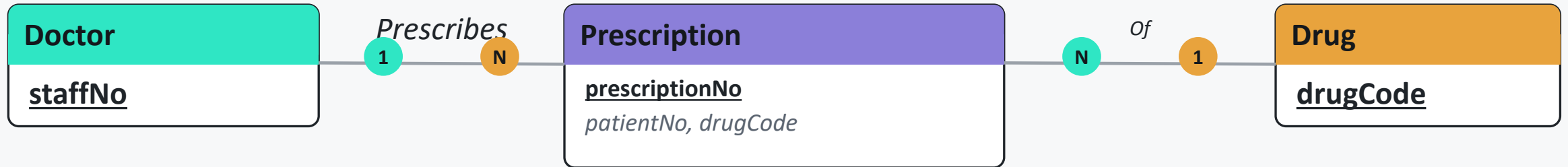
After — the relationship is an aggregate



WHY AGGREGATION HERE

The attending doctor belongs to this admission, not to the patient or the ward alone, so *AttendedBy* hangs off the aggregate. After mapping, attendingDoctorNo is an ordinary foreign key.

Prescriptions and Invoices



WHY PRESCRIPTION IS AN ENTITY

Prescription has its own dosage and schedule, and ties a patient, a doctor, and a drug together. Patient also has an Invoice per stay or visit (BilledTo, 1:N).

The Complete EER Model

- Patient is the hub: Books → Appointment, AdmittedTo → Admission, For → Prescription, BilledTo → Invoice.
- Admission is attended by a Doctor. Doctor sits inside the two-level Staff hierarchy.
- Bed is weak under Ward; Ward is composed under Department.
- Drawing all of it at once hides traps. Build by layer, check each against the brief.

Validating Against the Brief

Requirement	Covered by
Patients and their admissions	Patient; Admission (aggregate)
Ward, bed, admission and discharge	Admission attributes; weak Bed
Four staff roles with role data	Level 1 specialization
Consultants, residents, supervision	Level 2; supervisingConsultantNo
Department head, wards, beds	HeadedBy; ComposedOf; Owns
Attending doctor per admission	AttendedBy on Admission
Appointments with confirmation	Appointment; ScheduledWith
Prescriptions with dosage	Prescription entity
Invoices and payment status	Invoice; BilledTo

Assumptions to Confirm With the Client

1. A resident has exactly one supervising consultant

Implied by the brief, never stated. Never zero, never two.

2. Nurse and Technician have no further sub-roles

The brief gives no role detail beyond shift and labSpecialty. Confirm before treating it as final.

KEY TAKEAWAYS

Key Takeaways

- 1 A real design starts from client language. Entity and relationship tests do the sorting.
- 2 Specialization can nest: Staff → Doctor → {Consultant, Resident}, when a subclass needs more structure.
- 3 Weak entities, composition, and aggregation each answer a specific sentence in the brief.
- 4 Validate in both directions, and state every assumption to the client.